

Exercise 1

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Problem 1a. Let M be a R -module. Show that for every exact sequence of R modules $U \xrightarrow{f} V \xrightarrow{g} W \rightarrow 0$ the sequence

$$M \otimes U \xrightarrow{id \otimes f} M \otimes V \xrightarrow{id \otimes g} M \otimes W \longrightarrow 0$$

is exact.

Proof. Surjectivity of $id \otimes g$: $\forall y \in M \otimes W$, $y = \sum_{i=1}^n m_i \otimes w_i$ for $m_i \in M$ and $w_i \in W$. By surjectivity of g , $\forall w_i, \exists v_i \in V \ni g(v_i) = w_i$. Thus

$$(id \otimes g) \left(\sum_{i=1}^n m_i \otimes v_i \right) = \sum_{i=1}^n m_i \otimes g(v_i) = \sum_{i=1}^n m_i \otimes w_i = y$$

Thus $id \otimes g$ is surjective.

Exactness at $M \otimes V$: Let $h : \frac{M \otimes V}{\text{Im}(id \otimes f)} \rightarrow M \otimes W$ where $m \otimes v + \text{Im}(id \otimes f) \rightarrow m \otimes g(v)$. Well defined: $h(m_1 \otimes v + m_2 \otimes f(u) + \text{Im}(id \otimes f)) = m_1 \otimes g(v) + m_2 \otimes g(f(u)) = m_1 \otimes g(v)$ since $g(f(u)) = 0$ by exactness.

$\forall w \in W$, by surjectivity of $g \exists v \in V \ni g(v) = w$. Let $k : M \otimes W \rightarrow \frac{M \otimes V}{\text{Im}(id \otimes f)}$ where $\forall y \in M \otimes W$, $y = \sum_{i=1}^n m_i \otimes w_i \rightarrow \sum_{i=1}^n m_i \otimes v_i + \text{Im}(id \otimes f)$ where $g(v_i) = w_i$.

$\forall m \in M$ and $\forall v \in V$, $k(h(m \otimes v + \text{Im}(id \otimes f))) = k(m \otimes g(v)) = m \otimes v + \text{Im}(id \otimes f)$. By linearity of h and k , $k \circ h = id_{M \otimes V / \text{Im}(id \otimes f)}$. Thus h is injective. h inherits surjectivity from g . Thus g factors through h . Thus $\ker g = \text{Im}(id \otimes f)$. Thus sequence is exact at $M \otimes V$ $\square$